

Theory, Model, Equation, Teaching

A four-level account of the structure of physics and its pedagogy

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Abstract

Physics is commonly taught as a sequence of equations. This essay argues that equations become intelligible only when located within a four-level hierarchy. A **theory** supplies an architecture of configuration-kind and source-kind variables connected by structural, constitutive, dynamical and interaction equations (following Tonti). A **model** instantiates that architecture through selection of system and interactions, restriction to a constitutive regime, and specification of geometry and conditions. Each **equation** within the model carries both an architectural role and an epistemic status, and these two classifications are orthogonal: the same equation may shift from empirical law to identity under a change of representation. **Teaching** exploits the modularity by holding one part of the architecture fixed, a *hinge*, while students vary the case-specific completion. The synthesis locates the hinge precisely: the invariant module belongs to the theory, the completion belongs to the model, and the open slot identifies which modelling decision remains unmade.

The hierarchy

Four questions organise the account with each addressing a different object and being answered at a different level.

Level	Object	Governing question
Theory	Architecture	What kinds of quantities exist, and what roles do their equations play?
Model	Instantiation	Which selections, restrictions and specifications turn the theory into a solvable case?
Equation	Classification	Where does this equation sit, and what kind of claim is it making?
Teaching	Pedagogy	Which module can be held invariant while students vary the completion?

The logical order is theory $\rightarrow$ model $\rightarrow$ equation in context $\rightarrow$ teaching. A theory provides the space of admissible descriptions. Modelling selects and constrains one description. Equations acquire their concrete role and authority inside that model. Teaching then preserves one module while varying another so the architecture becomes visible.

Level 1: the grammar of a theory

Tonti's 1975 classification [1, 2, 3] partitions a theory's variables into configuration-kind (state, trajectory, potential) and source-kind (momentum, force, density, current), connected by constitutive equations and embedded within structural and interaction relations. The present account distinguishes four equation types by architectural role:

- **Structure equations** construct derived quantities within one family: $\mathbf{v} = d\mathbf{r}/dt$, $\mathbf{E} = -\nabla\phi$, $\mathbf{B} = \nabla \times \mathbf{A}$.
- **Constitutive equations** translate between families, carrying material or regime-dependent content: $\mathbf{p} = m\mathbf{v}$, $\mathbf{D} = \varepsilon\mathbf{E}$, $\mathbf{B} = \mu\mathbf{H}$.
- **Dynamical or balance equations** govern change: $d\mathbf{p}/dt = \mathbf{F}$, $\nabla \cdot \mathbf{D} = \rho_q$.
- **Interaction equations** couple one sector to another: $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$, $\mathbf{F} = m_g\mathbf{g}$.

Classical particle mechanics makes this vivid. The theory forms a loop:

$$\mathbf{r} \xrightarrow{\text{kinematics}} \mathbf{v} \xrightarrow{\text{constitution}} \mathbf{p} \xrightarrow{\text{dynamics}} \mathbf{F} \xrightarrow{\text{interaction}} \mathbf{r}.$$

The familiar compression $\mathbf{F} = m\mathbf{a}$ fuses three of these four modules, concealing the distinction between kinematic definition ($\mathbf{v} = \dot{\mathbf{r}}$), constitutive map ($\mathbf{p} = m\mathbf{v}$), and dynamical balance ($\dot{\mathbf{p}} = \mathbf{F}$). The loop closes only after an interaction law supplies the force; the symbol $\mathbf{F}$ is therefore a typed placeholder for a sum of interaction laws [4].

This architecture makes comparison between theories precise. The classical-to-relativistic transition replaces one constitutive cell ($\mathbf{p} = m\mathbf{v} \rightarrow \mathbf{p} = \gamma m_0\mathbf{v}$) while preserving the kinematic and dynamical slots. Newtonian gravity and electrostatics share a potential→field→flux→source sequence while differing in source type, sign structure and interaction law.

	Configuration side	Constitutive map	Source/interaction side
Classical mechanics	$\mathbf{r} \mapsto \mathbf{v}$	$\mathbf{p} = m\mathbf{v}$	$\dot{\mathbf{p}} = \mathbf{F}$, plus force law
Relativistic mechanics	$x^\mu \mapsto u^\mu$	$p^\mu = m_0 u^\mu$	$dp^\mu/d\tau = K^\mu$
Newtonian gravity	$V \mapsto \mathbf{g}$	$\mathbf{h} = -\mathbf{g}/(4\pi G)$	$\nabla \cdot \mathbf{h} = \rho_m$, $\mathbf{F} = m_g\mathbf{g}$
Electrostatics	$\phi \mapsto \mathbf{E}$	$\mathbf{D} = \varepsilon\mathbf{E}$	$\nabla \cdot \mathbf{D} = \rho_q$, $\mathbf{F} = q\mathbf{E}$
Magnetostatics	$\mathbf{A} \mapsto \mathbf{B}$	$\mathbf{B} = \mu\mathbf{H}$	$\nabla \times \mathbf{H} = \mathbf{J}$

Table 1: Five theories as variations within a common grammar.

A further observation concerns the theory boundary. Whether an interaction equation sits inside or outside the boundary depends on the chosen level of closure. A test-particle model treats the field as external; a self-consistent particle–field model enlarges the theory until both source generation and back-reaction are internal. An apparently incomplete diagram may therefore represent a controlled approximation rather than an error, [3]

Level 2: from theory to model

Physicists rarely solve theories in the abstract. They solve models constructed from theories. The construction requires decisions that are often compressed into phrases such as “assume,” “neglect,” “take” or “let.” These decisions fall into three functionally distinct groups [12, 13, 14]:

- **Selection** determines what the model contains: system boundary, retained degrees of freedom, included interactions, level of description.
- **Restriction** determines how the retained components may behave: constitutive law, idealisation, approximation, regime, symmetry.
- **Specification** determines which particular case is solved: geometry, parameters, coordinates, gauge, boundary conditions, initial data.

These three parses reduce different spaces of possibility. If $\mathcal{T}$ denotes the theory’s admissible architecture, selection identifies a sector $\mathcal{S} \subset \mathcal{T}$, restriction identifies a family $\mathcal{R} \subset \mathcal{S}$, and specification selects a determinate problem $M(\alpha_i, \mathcal{G}, \mathcal{B}, \mathcal{I})$.

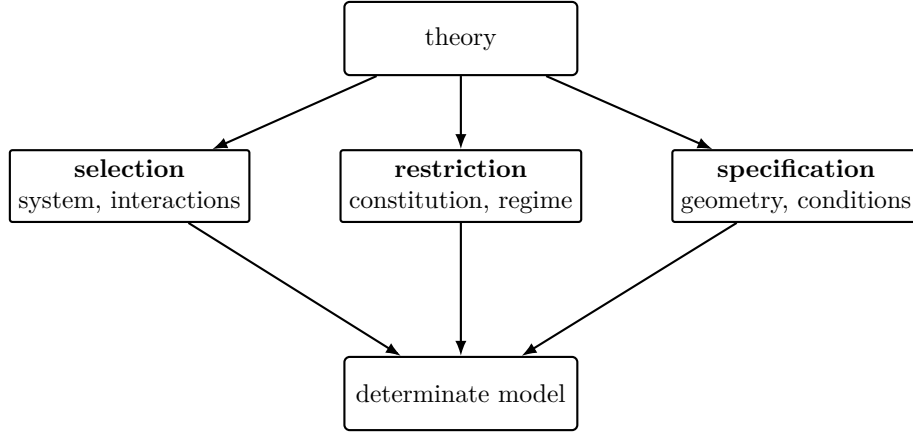


Figure 1: Three modelling moves, analytically distinct even when made together.

Consider a block on an inclined plane. Selection retains the block as focal body, translation as the degree of freedom, and gravity, normal reaction and friction as the included interactions. Restriction idealises the block as a point mass, the field as uniform $\mathbf{g}$, and the friction as Coulomb. Specification provides m , θ , μ_k , a coordinate axis parallel to the plane, and initial conditions. Only after all three parses does $F = ma$ become the solvable equation $\ddot{x} = g(\sin \theta - \mu_k \cos \theta)$. The cancellation of mass is a consequence of the completed model, not a free-standing property of Newton’s second law.

The distinction is analytically useful because each parse can fail independently. Omitting drag is a selection failure. Exceeding the small-angle regime is a restriction failure. Using the wrong coefficient of friction is a specification failure. “The model was wrong” is too coarse a diagnosis; the three-parse taxonomy locates where the model went wrong.

Morgan and Morrison describe models as partially autonomous instruments mediating between theories and the world [16]. Cartwright argues that fundamental laws achieve scope by abstracting from particular situations, and their application requires model-level constructions in which idealisations and constitutive closures are made operative [15]. The present framework formalises this mediation.

Level 3: the anatomy of an equation

Tonti’s grammar answers *where* an equation sits. A second question is equally important: *what kind of claim is it making?* These two classifications are orthogonal. An equation occupying a structural position may be a definition, an identity, or an empirical law. An equation occupying a constitutive position may be an exact relation, a linear approximation, or a regime-dependent model assumption.

The epistemic axis proposes the following working taxonomy, [19]:

Epistemic status	What the equation does
Definition	Introduces a quantity: $\mathbf{v} = d\mathbf{r}/dt$.
Identity / representation	True by mathematical structure: $\nabla \cdot (\nabla \times \mathbf{A}) = 0$.
Empirical law	Experimentally established: $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
Constitutive law	Material or regime-dependent: $\mathbf{D} = \varepsilon \mathbf{E}$.
Conservation / balance	Quantity conserved or balanced: $d\mathbf{p}/dt = \mathbf{F}$.
Approximation	Controlled simplification: $\sin \theta \approx \theta$.
Idealisation	Deliberately unreal: massless string, frictionless surface.
Boundary / initial condition	Supplementary data: $\phi \rightarrow 0$ as $r \rightarrow \infty$.
Axiom / postulate	Primitive constraint of a formulation.
Derived result	Consequence: $\nabla^2 \phi = -\rho_q/\varepsilon$.

The two axes form a matrix in which the same equation may shift position under a change of formulation. The equation $\nabla \cdot \mathbf{B} = 0$ illustrates this cleanly. Given $\mathbf{B} = \nabla \times \mathbf{A}$, the vanishing divergence follows as a vector-calculus identity. Yet the physical availability of a global vector potential rests on the empirical absence of magnetic monopoles; Dirac showed that even a single monopole would quantise electric charge [17]. The architectural role (structural constraint) is stable; the epistemic status (identity or empirical law) depends on whether the equation is read as a consequence of a representation or as a physical assertion that the representation is available.

$\mathbf{F} = m\mathbf{a}$ is epistemically ambiguous in a different way: it can be read as a definition of force (Mach), a definition of inertial mass (operational reading), an empirical law (given independent definitions), or a constitutive-dynamical fusion (the loop decomposition). Newton himself used it as two steps: defining “quantity of motion” as $m\mathbf{v}$, then relating impressed force to the change of motion. The compression into $\mathbf{F} = m\mathbf{a}$ is a later convenience that obscures the original structure.

$pV = nRT$ shows a single equation shifting epistemic status while its architectural role (constitutive) remains fixed: it is an empirical law for dilute gases, an idealisation for real gases, and a derived result in statistical mechanics where temperature is defined through $\partial S/\partial E$.

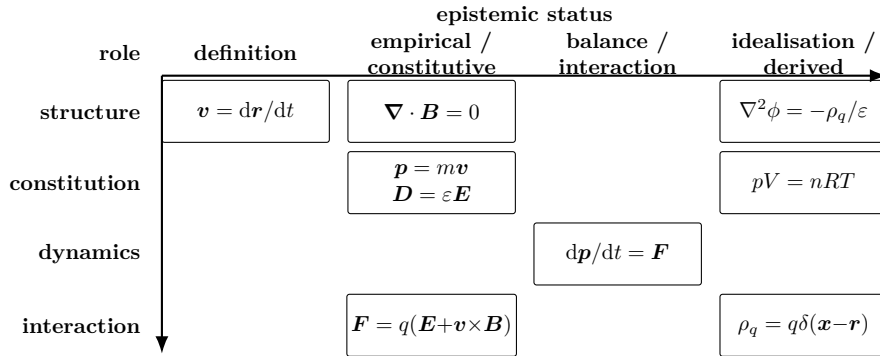


Figure 2: The two-axis classification. An equation may move horizontally (epistemic shift) or vertically (architectural reassignment under reformulation).

Equations routinely shift epistemic status through five mechanisms: representation adoption ($\nabla \cdot \mathbf{B} = 0$ becomes an identity after introducing $\mathbf{A}$); theory change ($\mathbf{p} = m\mathbf{v}$ becomes an approximation in relativistic mechanics); change of primitive ($pV = nRT$ becomes a derived result when T is defined statistically); scope change (ideal-gas law becomes an idealisation at high pressure); and pedagogical framing (the same equation may be presented as a definition or an empirical law depending on which quantities have been established first) [10].

Level 4: the hinge

The modularity exposed at levels 1–3 explains a recurring pedagogical phenomenon. Many familiar equations function as *hinges*: they organise reasoning while leaving a meaningful slot open. The hinge principle is:

$$\boxed{\text{shared relation}} + \boxed{\text{case-specific closure}} + \boxed{\text{system and assumptions}} \longrightarrow \boxed{\text{physical account}}.$$

The dynamical balance $d\mathbf{p}/dt = \mathbf{F}$ organises the relation between momentum change and force without specifying the physical origin of $\mathbf{F}$. The cases $\mathbf{F} = m\mathbf{g}$, $\mathbf{F} = -k\mathbf{x}$, $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ supply distinct interaction closures. The wave relation $v = f\lambda$ organises three kinematic quantities without specifying the wave speed; a medium model such as $v = \sqrt{T/\mu}$ or $v = 1/\sqrt{\epsilon\mu}$ completes it. The first law $\Delta U = Q - W$ organises the energy account without specifying the equation of state, process path or system boundary.

Hinge	Shared relation	Cases	Closure
Dynamics	$F_{\text{net}} = ma$	gravity; spring; Lorentz	interaction law
Momentum	$F = \Delta p/\Delta t$	collision; rocket; gas pressure	impulse model
Waves	$v = f\lambda$	string; sound; light	medium law
Power	$P = \Delta E/\Delta t$	lifting; resistor; lamp	transfer pathway
Energy	$\Delta U = Q - W$	cycle; Kirchhoff; nuclear	equation of state, process
Flux	total/area	pressure; irradiance; Φ_B	distributed quantity

Table 2: Selected hinge structures across IB Physics.

The hinge works pedagogically because the invariant module and the case-specific completion occupy different positions on both axes. The hinge relation is typically definitional or structural (high epistemic authority, low physical specificity); the completion is typically empirical or constitutive (regime-dependent authority, high physical specificity). Students who identify this asymmetry are doing what Sherin calls activating a *symbolic form* [6]: recognising that the equation has an internal structure in which different parts carry different kinds of authority. Chi, Feltovich and Glaser showed that experts sort problems by principle while novices sort by surface features [5]; the hinge framework trains the expert sort.

The three-parse modelling taxonomy adds a further refinement. The hinge’s open slot is an unmade modelling decision, and the kind of completion required identifies which parse remains open:

Open slot	Modelling parse	Unanswered question
Interaction law	Selection	Which coupling is included?
Constitutive/regime	Restriction	How may the system respond?
Geometry/conditions	Specification	Which particular case?

The student who identifies the open slot is therefore diagnosing which modelling decision remains unmade. This is closer to professional physical reasoning than either “find the right equation” or “understand the concept” treated in isolation.

What the hierarchy clarifies

The four levels prevent several common conflations.

Theory and model. A theory supplies admissible architecture; a model instantiates it through specific choices. $d\mathbf{p}/dt = \mathbf{F}$ belongs to the theory; $\ddot{x} = g(\sin\theta - \mu_k \cos\theta)$ belongs to a model constructed from that theory through selection, restriction and specification. Saying “Newton’s second law predicts that the block accelerates at $g \sin\theta$ ” conflates theory with model, omitting the friction decision, the coordinate choice, and the idealisation of uniform g .

Architectural role and epistemic status. $\mathbf{D} = \varepsilon\mathbf{E}$ and $\mathbf{p} = m\mathbf{v}$ both occupy constitutive positions, but the first is a material model carrying linear and isotropic assumptions while the second is a regime-dependent law replaced wholesale under relativity. Treating both as “just constitutive” misses the difference in what each claims about the world.

Compression and structure. $\mathbf{F} = m\mathbf{a}$ fuses kinematics, constitution and dynamics; $\nabla^2\phi = -\rho_q/\varepsilon$ fuses structure, constitution and source. Compression is efficient after the architecture is understood; before then, it encourages the treatment of equations as self-interpreting templates. Clement’s reversal-error studies [8] and Torigoe’s unpacking research [9] document the consequences: students read compressed equations as sentences rather than as structured representations.

Physical change and representational change. Choosing a gauge ($\nabla \cdot \mathbf{A} = 0$) or a coordinate system (axes parallel to an incline) changes the equations without changing the physics. A student who treats a gauge condition as another empirical field equation is confusing a specification move with an empirical law. The two-axis classification exposes the error: the modelling function is specification, while the epistemic status is representational convention.

Implications for teaching

A hinge lesson follows six moves: state the shared relation and its status; expose the open slot; compare three cases with distinct closures; name each closure by type (interaction law, constitutive relation, equation of state, geometry, conservation condition, approximation); interrogate signs, direction and domain of validity; synthesise by contrasting the invariant structure against the varying physical closure and resulting observable.

Assessment credits structure as well as calculation, [18]. A four-feature scheme allocates marks to: shared structure (correct relation, variables, conditions), physical closure (correct case-specific law or constraint), system and assumptions (boundary, direction, controlled variable, approximation), and interpretation (clear statement of what follows physically). A student who manipulates the shared equation correctly while choosing an unsuitable closure, or who knows the local law while misidentifying the system, receives diagnostic rather than merely numerical feedback. The recurring classroom question is concise:

What does this equation organise, and what physics completes the case?

Conclusion

A formula sheet presents equations as if they were all the same kind of thing. They are not. Definitions, identities, empirical laws, constitutive assumptions, interaction couplings, boundary conditions and idealisations differ in origin, in authority, in domain of validity, and in what happens when they fail. The four-level hierarchy makes those differences visible. A theory supplies an architecture. A model instantiates that architecture through selection, restriction and

specification. Each equation within the model carries both an architectural role and an epistemic status. Teaching exploits the modularity by holding one module fixed, the hinge, while students identify the completion, which is simultaneously a case-specific equation and an unmade modelling decision. The student who can distinguish the invariant architecture from the case-specific closure, the structural from the constitutive, the definition from the empirical law, and the selection from the restriction, has access to the grammar that makes physics transferable. That grammar is what a formula sheet conceals and what the hierarchy restores, [3, 18, 19, 14].

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